

To study the shift in equilibrium between ferric ions and thiocyanate ions by increasing/ decreasing the concentration of these ions

Video Reference :- <https://www.youtube.com/watch?v=zCeciBrBhMU>

Materials Required :-

- 4 Test Tubes
- Test Tube Stand
- Distilled Water
- Burette
- Cobalt Chloride Solution
- Aceton
- Concentrated Hydrochloric Acid
- Measuring Cylinder

Procedure :-

1. Study Of Equilibrium Shift When Adding Water

- Pour 3 ml of Cobalt Chloride Solution in 5 test tubes (A, B, C, D, E)
- Pour 1 ml of Acetone Solution in the A test tube.
- Pour 0.8 ml of Acetone Solution in the B test tube.
- Pour 0.6 ml of Acetone Solution in the C test tube.
- Pour 0.4 ml of Acetone Solution in the D test tube.
- Pour 0.2 ml of Acetone Solution in the E test tube.
- Pour 0.2 ml of Distilled Water in the B test tube.
- Pour 0.4 ml of Distilled Water in the C test tube.
- Pour 0.6 ml of Distilled Water in the D test tube.
- Pour 0.8 ml of Distilled Water in the E test tube.

1. Study Of Equilibrium Shift When Adding Cl⁻ ions

- Pour 10 ml of Cobalt Chloride Solution in X test tube.
- Add 10 ml of distilled water in X test tube
- Pour 1.5 ml of the solution from X test tube to A test tube.
- Pour 1.5 ml of the solution from X test tube to B test tube.

- Pour 1.5 ml of the solution from X test tube to C test tube.
- Pour 1.5 ml of the solution from X test tube to D test tube.
- Pour 1.5 ml of the solution from X test tube to E test tube.
- Pour 2 ml of Distilled Water in the A test tube.
- Pour 1.5 ml of Distilled Water in the B test tube.
- Pour 1 ml of Distilled Water in the C test tube.
- Pour 0.5 ml of Distilled Water in the D test tube.
- Pour 0.5 ml of Concentrated Hydrochloric Acid in the A test tube.
- Pour 1 ml of Concentrated Hydrochloric Acid in the B test tube.
- Pour 1.5 ml of Concentrated Hydrochloric Acid in the C test tube.
- Pour 2 ml of Concentrated Hydrochloric Acid in the D test tube.
- Pour 2.5 ml of Concentrated Hydrochloric Acid in the E test tube.